Xi'an Jiaotong-Liverpool University

西交利物浦大学

PAPER CODE	EXAMINER	DEPARTMENT	TEL
INT305		Intelligent Science	

1st SEMESTER 2024/25 Final Exam

BACHELOR DEGREE – Year 4

Machine Learning

TIME ALLOWED: 3 Hours 0 Minutes

INSTRUCTIONS TO CANDIDATES

1、 Total marks available are 100.

2、 Answer all questions.

3、 The number in the column on the right indicates the approximate marks for each section.

4、 In general, it is particularly important to give reasons for your answer. Only partial marks will be awarded for correct answers with inadequate reasons.

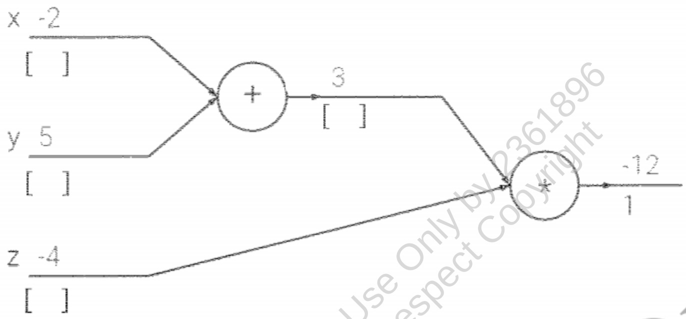
5、 Answer should be written in the answer booklet(s) provided.

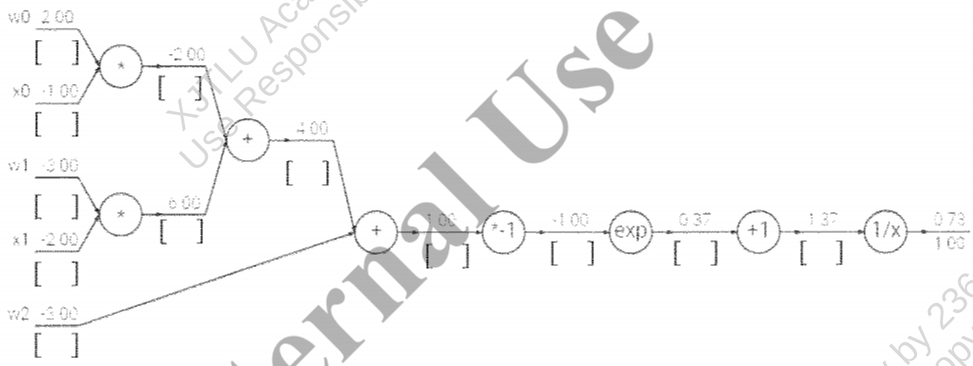
6、 The university approved calculator - Casio FS82ES/83ES can be used.

7、 Only solutions written in English will be accepted.

Answer All Questions

1. Fill in the missing gradients underneath the forward pass activations in each circuit diagram. The gradient of the output with respect to the loss is one (1.00), and has already been filled in. (The output values for each unit are represented on top of the line and the gradients are below the line.)
Please calculate the corresponding gradient **below** each unit in the brackets.

1) 

2) 

Total 16 (1 mark for each blank)

2. The Zero-One Loss is a simple classification loss function that intuitively counts how many mistakes the model makes. In this problem we will examine the Zero-One Loss in the context of binary classification, where the loss for a single example x_i with label $y_i \in \{0,1\}$ is given by:

$$L(x_i) = \mathbb{1}[s_{y_i} - s_{(1-y_i)} < 0]$$

where s_j represents the model's predicted score for the $j-th$ class and is an indicator function that is 1 if the condition inside the function is true, and 0 otherwise.

1) How to derive the critical point of 0-1 loss? **(4')**
2) Is it a good idea to use 0-1 loss for the gradient descent optimization process? Why? **(8')**

Total 12

3. You are given a dataset that includes attributes Hours of Sleep (Low, Medium, High), whether the person exercised (Exercised: Yes/No), and whether they performed well in an exam (Performance: Good/Bad). Using this dataset, construct a decision tree to predict whether a person performs well. Answer the following:

Hours of Sleep	Exercised	Performance
Low	No	Bad
Low	Yes	Bad
Medium	No	Bad
Medium	Yes	Good
High	No	Good
High	Yes	Good

1) What is the entropy $H(Performance)$? **(4')**
2) What is the entropy $H(Performance\ |Hoursof\ Sleep)$? **(4')**
3) What is the entropy $H(Performance\ |\ Exercised)$? **(4')**
4) Construct the optimal decision tree. Explain why your decision tree is optimal. **(4')**

Total 16

4. Convolution operation is defined as follows:

$$f[x,y] * g[x,y] = \sum_{n_1=-\infty}^{\infty} \sum_{n_2=-\infty}^{\infty} f[n_1,n_2] \cdot g[x-n_1, y-n_2]$$

The image patch is defined as follows

$$f = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 3 & 1 & 2 & 1 \\ 0 & 2 & 1 & 3 \\ 1 & 0 & 2 & 1 \end{pmatrix}$$

The convolution filter is defined as

$$g = \begin{pmatrix} 1 & 0 & -1 \\ 0 & -1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

1) Perform a valid convolution operation on the input matrix f using the filter g with a stride of 1, no padding in this case. Provide the final output matrix. **(6')**

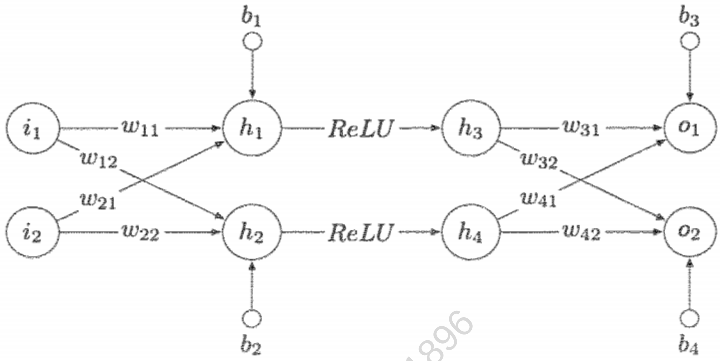
2) Perform a same convolution operation using the same input matrix f using the filter g, zero-padding with 1. Provide the padded input and the corresponding resulting output matrix. **(10')**

3) Explain how changing the stride affects the size of the output matrix. If the stride is increased to 2 for the same filter and the padded input in 2), what will the size of the output matrix be? **(4')**

Total 20

5. Given the following neural network with fully connected layer and ReLU activations, including two input units (i_1, i_2), four hidden units (h_1, h_2) and (h_3, h_4). The output units are indicated as (o_1, o_2) and their

targets are indicated as $(t_1,\ t_2)$. The weights and bias of fully connected layer are called w and b with specific sub-descriptors.



(hints: $h_1 = [w_{11}\ w_{21}]\begin{bmatrix} i_1 \\ i_2 \end{bmatrix} + b_1$)

The values of variables are given in the following table:

Variable	i_1	i_2	w_{11}	w_{12}	w_{21}	w_{22}	w_{31}	w_{32}	w_{41}	w_{42}	b_1	b_2	b_3	b_4	t_1	t_2
Value	2.0	-1.0	1.0	-0.5	0.5	-1.0	0.5	-1.0	-0.5	1.0	0.5	-0.5	-1.0	0.5	1.0	0.5

1) List the parameters in this network. **(2')**

2) Compute the output $(o_1,\ o_2)$ with the input $(i_1,\ i_2)$ and network parameters as specified above. Write down all calculations, including intermediate layer results. **(4')**

3) Compute the Squared error loss of the output $(o_1,\ o_2)$ calculated above and the target $(t_1,\ t_2)$. **(4')**

4) Update the weight w_{21} using gradient descent with learning rate 0.1 as well as the loss computed previously. (Please write down all your computations.) **(6')**

Total 16

6. We will use Gaussian Mixture Models (GMM) to perform clustering on a given dataset $\mathcal{D}$.

Assume that a data point $\mathbf{x}$ is generated through the following process:

$$p(z=k) = \pi_k$$
$$p(\mathbf{x}|z=k) = \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_k, \mathbf{I})$$

This defines joint distribution $p(z, \mathbf{x}) = p(z)p(\mathbf{x}|z)$ with parameters $\{\pi_k, \boldsymbol{\mu}_k\}_{k=1}^{K}$.

1) How to derive the probability distribution of $p(x)$? **(5')**

2) What is the log-like function to be maximized for this GMM? **(5')**

3) How to optimize the log-like function derived in 2). **(10')**

Total 20

The end of the paper

Xi'an Jiaotong-Liverpool University

西交利物浦大学

PAPER CODE	EXAMINER	DEPARTMENT	TEL
INT305	Siyue Yu	Intelligent Science	1507

1st SEMESTER 2024/25 Final Exam

BACHELOR DEGREE – Year 4

Machine Learning

TIME ALLOWED: 3 Hours 0 Minutes

INSTRUCTIONS TO CANDIDATES

1、 Total marks available are 100.

2、 Answer all questions.

3、 The number in the column on the right indicates the approximate marks for each section.

4、 In general, it is particularly important to give reasons for your answer. Only partial marks will be awarded for correct answers with inadequate reasons.

5、 Answer should be written in the answer booklet(s) provided.

6、 The university approved calculator - Casio FS82ES/83ES can be used.

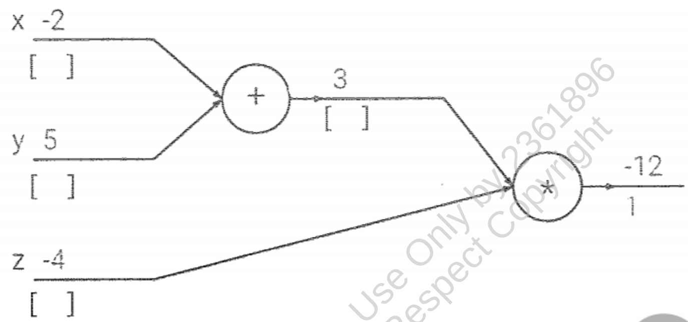
7、 Only solutions written in English will be accepted.

Answer All Questions

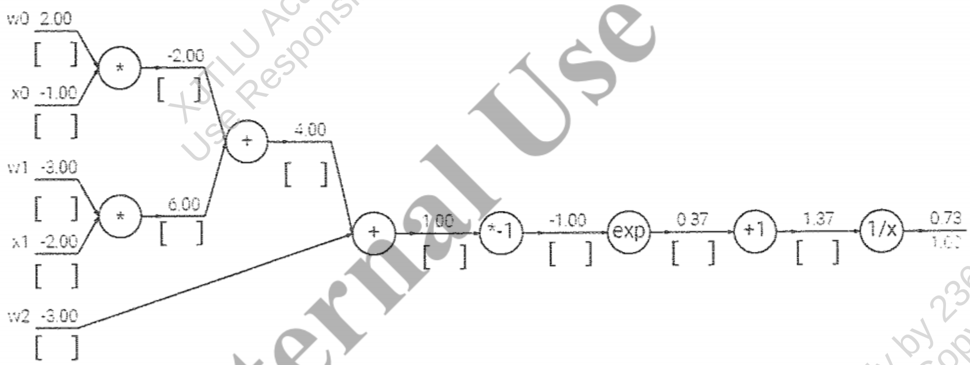
1. Fill in the missing gradients underneath the forward pass activations in each circuit diagram. The gradient of the output with respect to the loss is one (1.00), and has already been filled in. (The output values for each unit are represented on top of the line and the gradients are below the line.)

Please calculate the corresponding gradient **below** each unit in the brackets.

1)



2)



Total
16
(1 mark for each blank)

2. The Zero-One Loss is a simple classification loss function that intuitively counts how many mistakes the model makes. In this problem we will examine the Zero-One Loss in the context of binary classification, where the loss for a single example x_i with label $y_i \in \{0,1\}$ is given by:

$$L(x_i) = \mathbb{1}[s_{y_i} - s_{(1-y_i)} < 0]$$

where s_j represents the model's predicted score for the $j-th$ class and is an indicator function that is 1 if the condition inside the function is true, and 0 otherwise.

1) How to derive the critical point of 0-1 loss? **(4')**
2) Is it a good idea to use 0-1 loss for the gradient descent optimization process? Why? **(8')**

Total
12

3. You are given a dataset that includes attributes Hours of Sleep (Low, Medium, High), whether the person exercised (Exercised: Yes/No), and whether they performed well in an exam (Performance: Good/Bad). Using this dataset, construct a decision tree to predict whether a person performs well. Answer the following:

Hours of Sleep	Exercised	Performance
Low	No	Bad
Low	Yes	Bad
Medium	No	Bad
Medium	Yes	Good
High	No	Good
High	Yes	Good

1) What is the entropy $H(Performance)$? **(4')**
2) What is the entropy $H(Performance\ |Hoursof\ Sleep)$? **(4')**
3) What is the entropy $H(Performance\ |\ Exercised)$? **(4')**
4) Construct the optimal decision tree. Explain why your decision tree is optimal. **(4')**

Total 16

4. Convolution operation is defined as follows:

$$f[x,y]*g[x,y]=\sum_{n_1=-\infty}^{\infty}\sum_{n_2=-\infty}^{\infty}f[n_1,n_2]\cdot g[x-n_1,y-n_2]$$

The image patch is defined as follows

$$f=\begin{pmatrix}1&2&0&1\\3&1&2&1\\0&2&1&3\\1&0&2&1\end{pmatrix}$$

The convolution filter is defined as

$$g=\begin{pmatrix}1&0&-1\\0&-1&0\\1&0&1\end{pmatrix}$$

1) Perform a valid convolution operation on the input matrix f using the filter g with a stride of 1, no padding in this case. Provide the final output matrix. **(6')**

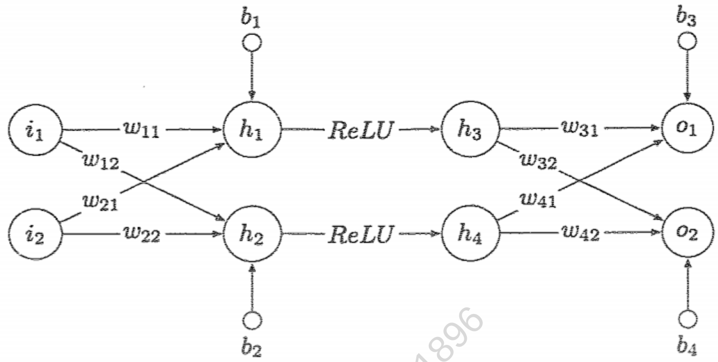
2) Perform a same convolution operation using the same input matrix f using the filter g, zero-padding with 1. Provide the padded input and the corresponding resulting output matrix. **(10')**

3) Explain how changing the stride affects the size of the output matrix. If the stride is increased to 2 for the same filter and the padded input in 2), what will the size of the output matrix be? **(4')**

Total 20

5. Given the following neural network with fully connected layer and ReLU activations, including two input units (i_1, i_2), four hidden units (h_1, h_2) and (h_3, h_4). The output units are indicated as (o_1, o_2) and their

targets are indicated as $(t_1,\ t_2)$. The weights and bias of fully connected layer are called w and b with specific sub-descriptors.



(hints: $h_1 = [w_{11}\ w_{21}]\begin{bmatrix} i_1 \\ i_2 \end{bmatrix} + b_1$)

The values of variables are given in the following table:

Variable	i_1	i_2	w_{11}	w_{12}	w_{21}	w_{22}	w_{31}	w_{32}	w_{41}	w_{42}	b_1	b_2	b_3	b_4	t_1	t_2
Value	2.0	-1.0	1.0	-0.5	0.5	-1.0	0.5	-1.0	-0.5	1.0	0.5	-0.5	-1.0	0.5	1.0	0.5

1) List the parameters in this network. **(2')**

2) Compute the output $(o_1,\ o_2)$ with the input $(i_1,\ i_2)$ and network parameters as specified above. Write down all calculations, including intermediate layer results. **(4')**

3) Compute the Squared error loss of the output $(o_1,\ o_2)$ calculated above and the target $(t_1,\ t_2)$. **(4')**

4) Update the weight w_{21} using gradient descent with learning rate 0.1 as well as the loss computed previously. (Please write down all your computations.) **(6')**

Total 16

6. We will use Gaussian Mixture Models (GMM) to perform clustering on a given dataset $\mathcal{D}$.

Assume that a data point $\mathbf{x}$ is generated through the following process:

$$p(z=k)=\pi_k$$
$$p(\mathbf{x}|z=k)=\mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_k,\mathbf{I})$$

This defines joint distribution $p(z,\,\mathbf{x}) \;=\; p(z)p(\mathbf{x}|z)$ with parameters $\{\pi_k,\,\boldsymbol{\mu}_k\}_{k=1}^{K}$.

1) How to derive the probability distribution of $p(x)$? **(5')**

2) What is the log-like function to be maximized for this GMM? **(5')**

3) How to optimize the log-like function derived in 2). **(10')**

Total 20

The end of the paper